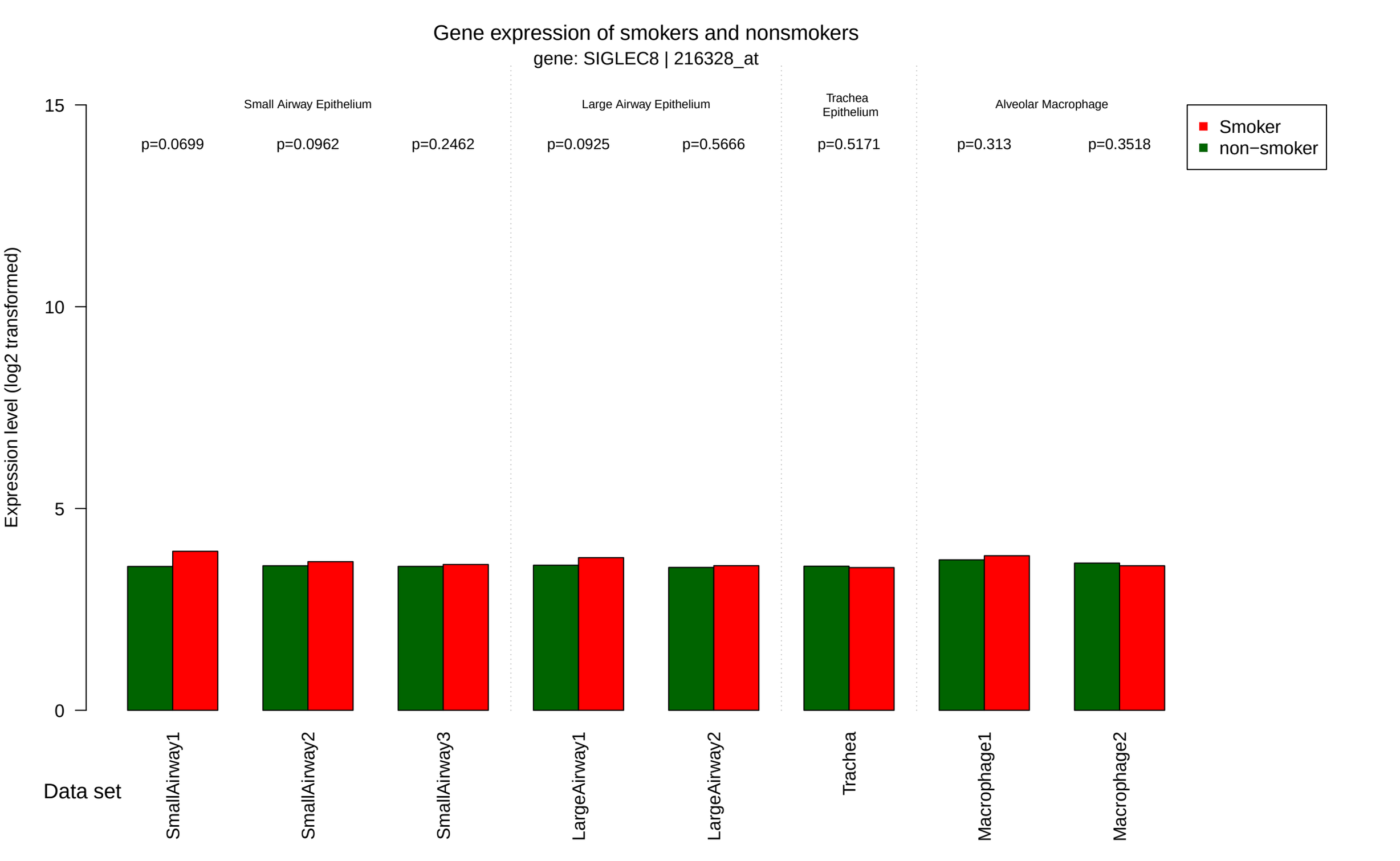
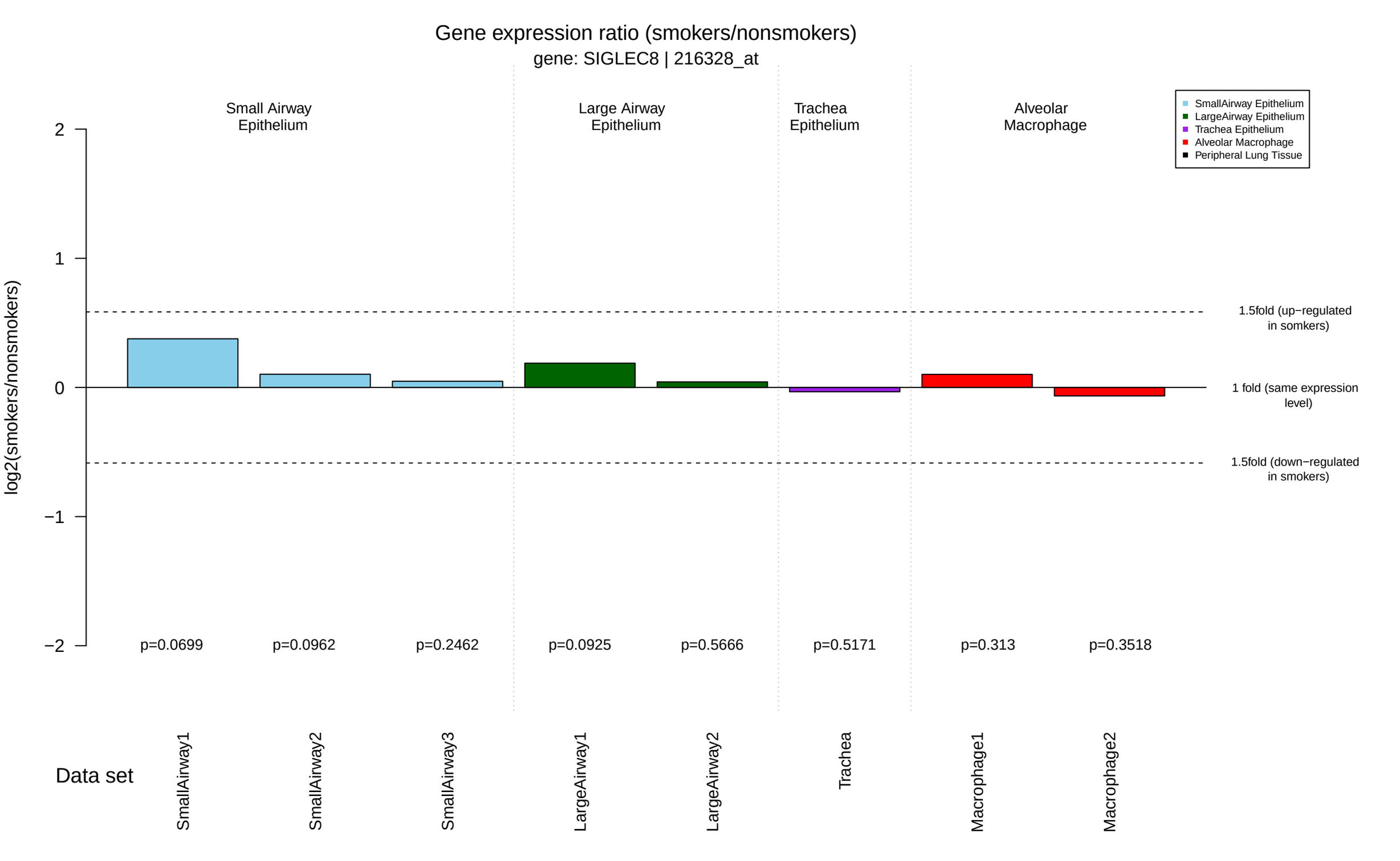
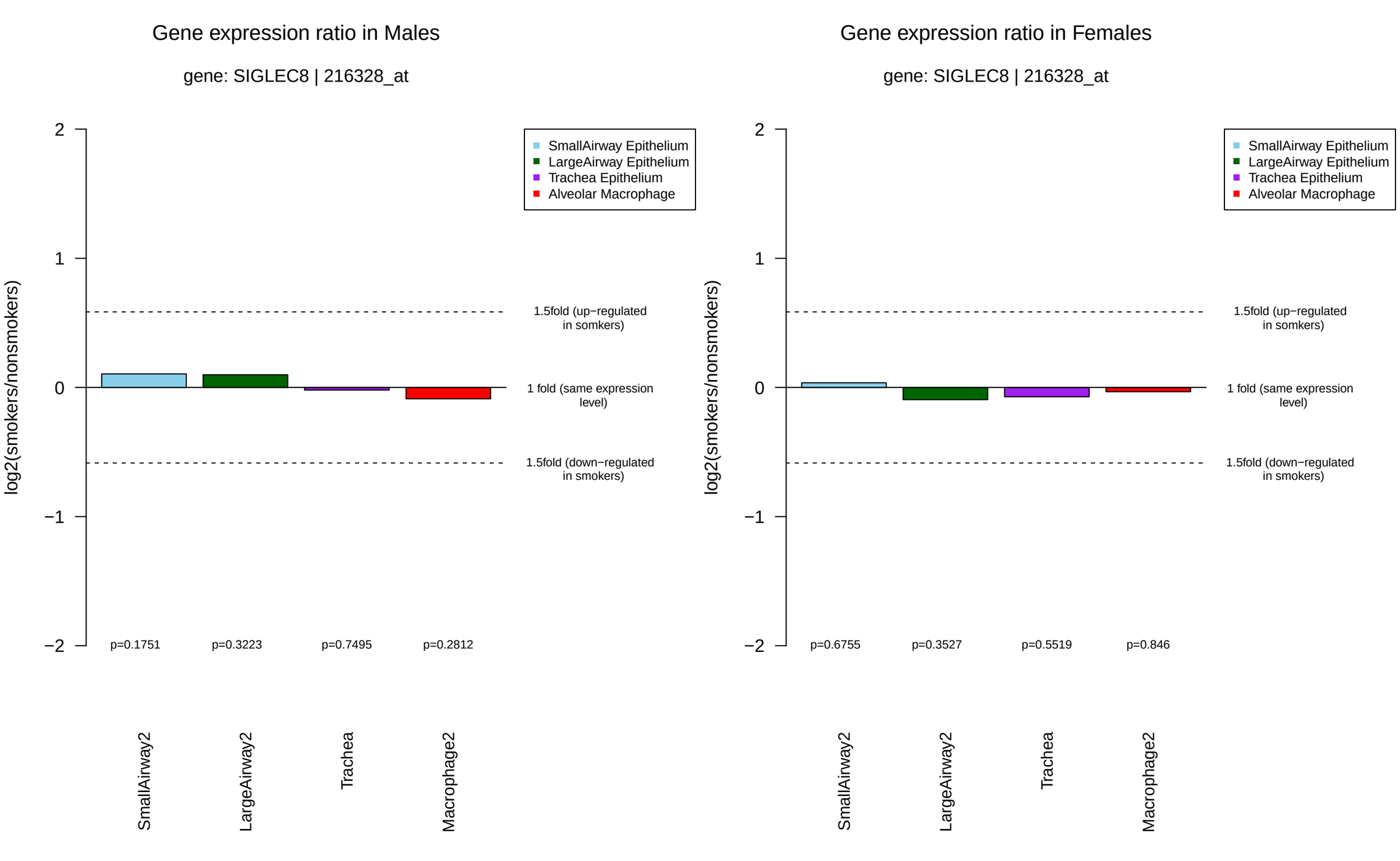
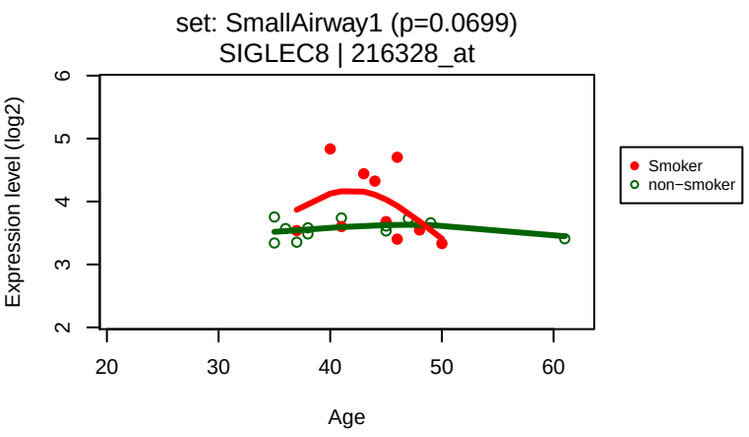
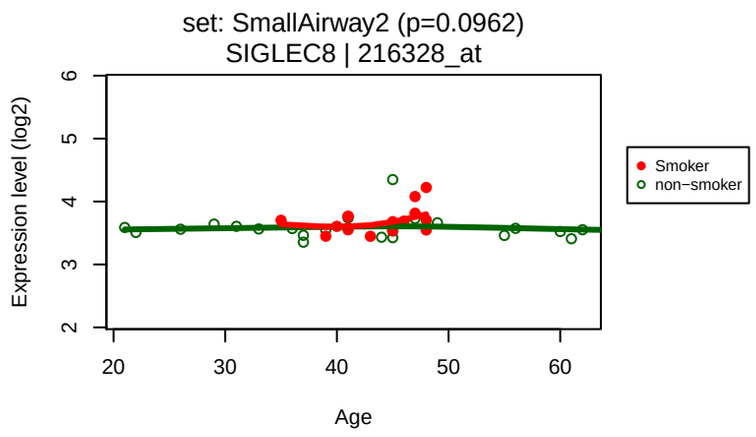
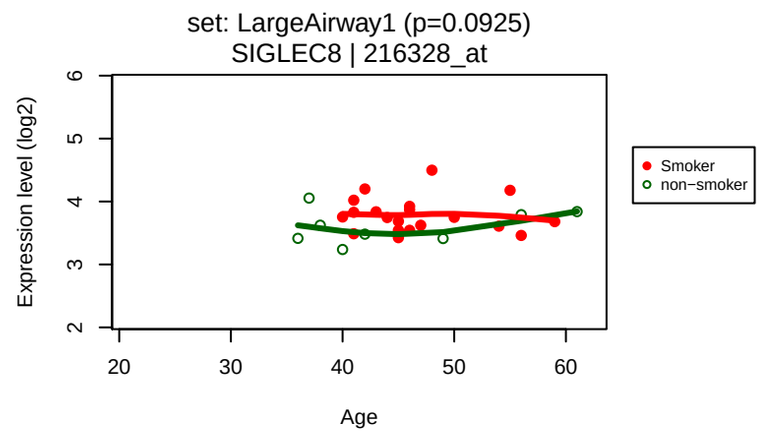

Gene expression of smokers and nonsmokers
gene: SIGLEC8 | 216328_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0699
p=0.0962
p=0.2462
p=0.0925
p=0.5666
p=0.5171
p=0.313
p=0.3518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

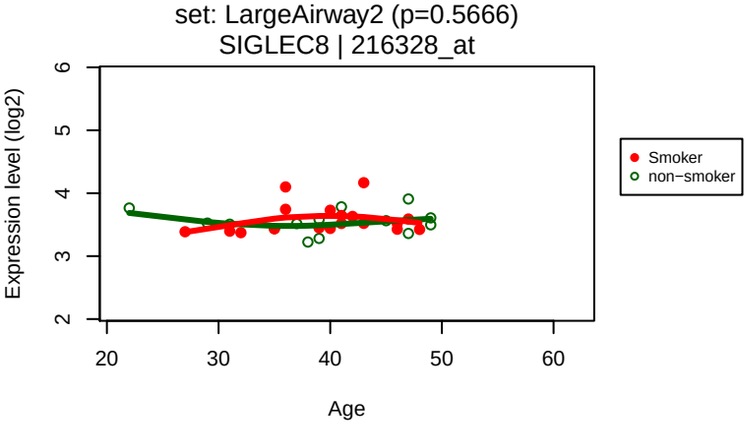

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC8 | 216328_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0699
p=0.0962
p=0.2462
p=0.0925
p=0.5666
p=0.5171
p=0.313
p=0.3518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

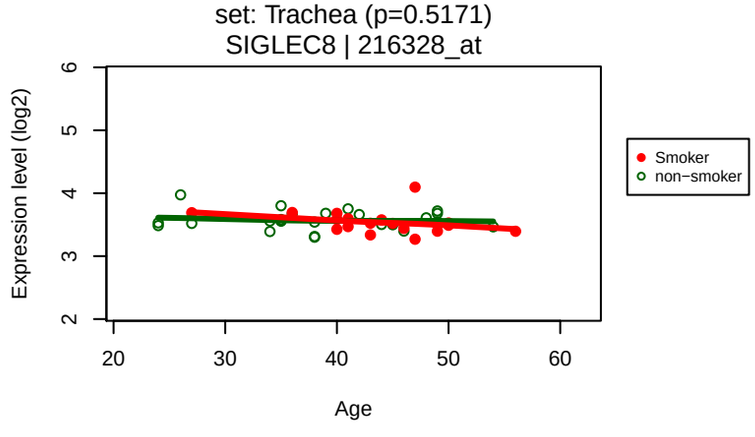

Gene expression ratio in Males
gene: SIGLEC8 | 216328_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1751
p=0.3223
p=0.7495
p=0.2812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC8 | 216328_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6755
p=0.3527
p=0.5519
p=0.846
SmallAirway2
LargeAirway2
Trachea
Macrophage2

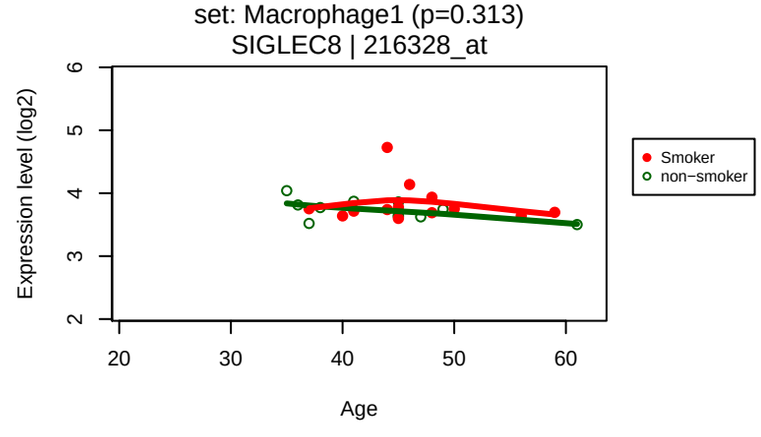

set: SmallAirway1 (p=0.0699)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker

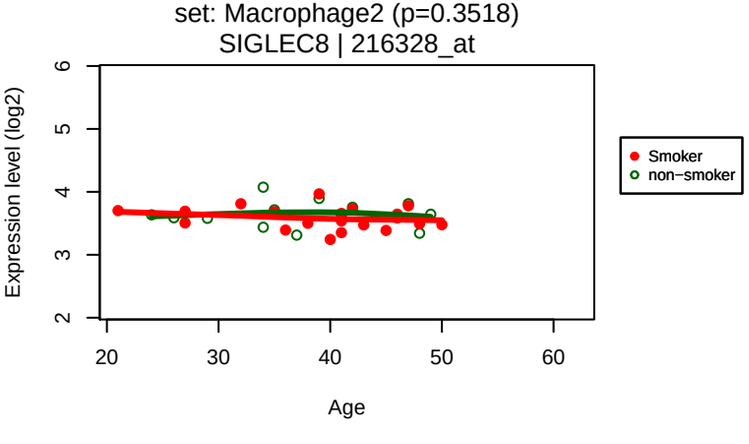

set: SmallAirway2 (p=0.0962)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker

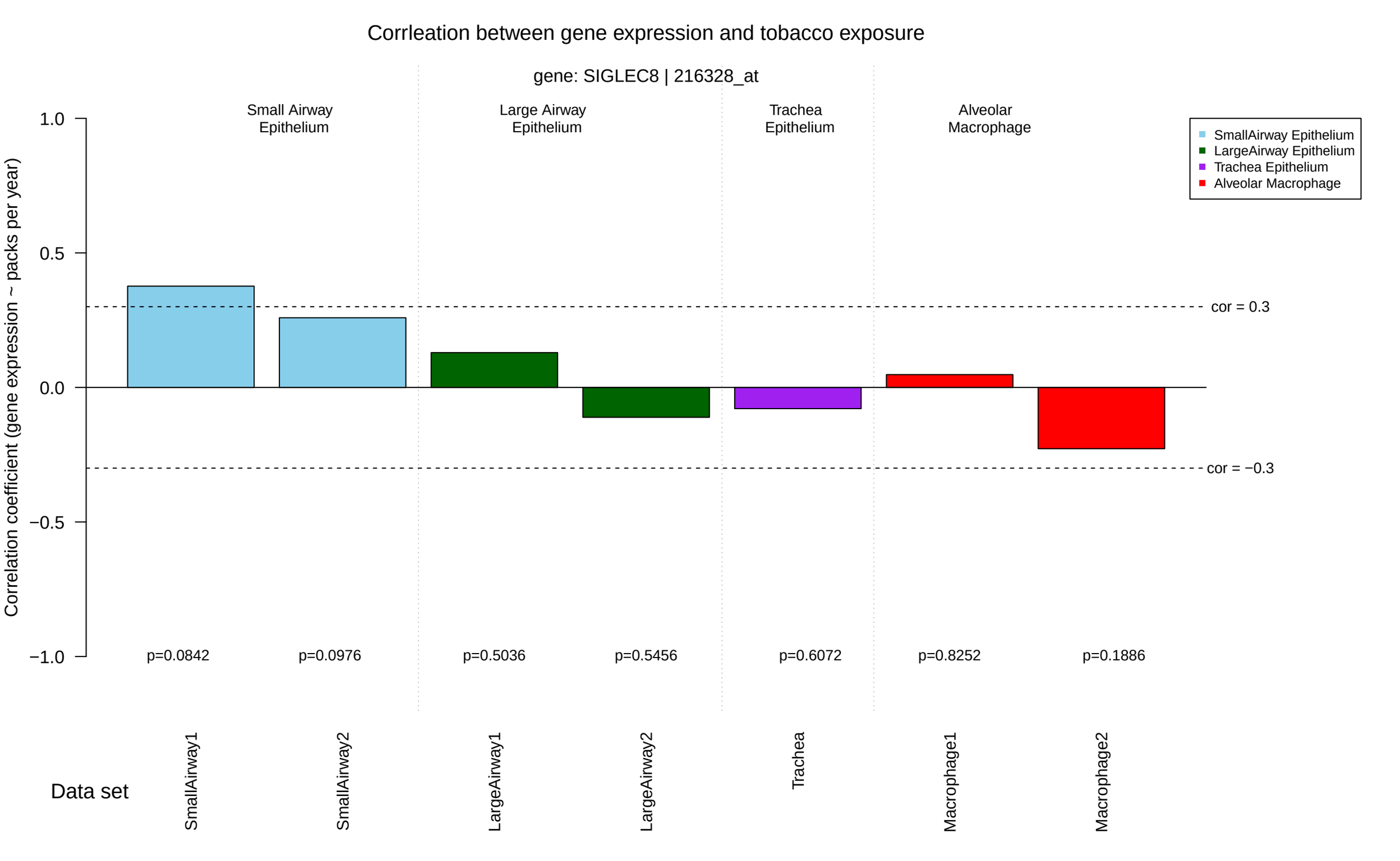

set: LargeAirway1 (p=0.0925)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5666)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5171)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.313)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3518)
SIGLEC8 | 216328_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC8 | 216328_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0842
p=0.0976
p=0.5036
p=0.5456
p=0.6072
p=0.8252
p=0.1886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2